R&D Data Problem – Senior Analyst

Data Summary

The attached file contains a dataset of pitches thrown by 200 different pitchers against 200 different batters. It contains the following columns:

- batter_id: numeric code identifying the batter (1-200)
- pitcher_id: numeric code identifying the pitcher (1-200)
- throw_side: pitcher's throwing hand (either L or R)
- bat_side: side of the plate batter hits from (either L or R)
- is_fb: binary indicator for whether the pitch is a Fastball
- is_in_zone: binary indicator for whether the pitch was thrown in the zone
- is_swing: binary indicator for whether the pitch was swung at by the batter
- is_miss: binary indicator for whether the pitch was swung at and missed

Prompt

R&D leadership has asked you to estimate the "true" contact ability of hitters throughout the MLB. That is, at what rate should we expect each hitter to make contact with a pitch when they swing. Their ask comes in three parts:

- What should we expect each hitter's contact rate to be going forward against all pitches thrown in the zone?
- What should we expect the contact rates to be going forward against all pitches thrown in the zone by left-handed pitchers?
- What should we expect the contact rates to be going forward against all fastballs thrown in the zone by left-handed pitchers?

R&D Problem Set

Deliverable

Please provide a writeup of your work along with any code used to achieve the work. This writeup can contain graphics to support your analysis if you'd like. Please also include answers to the following questions:

- What other information might you have wanted to support your analysis?
- What is one other interesting question you might want to explore with this dataset?